

Notes: Diamond (JPE, 1991)
**Monitoring and Reputation: The Choice between Bank Loans and Directly
Placed Debt**

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1 Big picture

- **Question.** When should a borrower use monitored bank debt rather than direct debt such as bonds or commercial paper?
- **Central friction.** Borrowers have private type and may face a moral-hazard choice between a safe positive-NPV project and a risky negative-NPV project.
- **Role of reputation.** A repayment track record changes lenders' beliefs and gives borrowers future rents to protect.
- **Role of monitoring.** A bank can spend resources to monitor whether a borrower chooses the risky action. Monitoring can provide incentives or merely screen out borrowers who are caught.
- **Main result.** High-rated borrowers borrow directly without monitoring; middle-rated borrowers rely on bank loans; very low-rated borrowers may be screened or receive no credit.
- **Life-cycle result.** New borrowers often begin with monitored bank loans, build a good record, and later graduate to direct debt once reputation alone provides discipline.
- **Aggregate implication.** High real interest rates or low expected future profitability reduce reputation rents and push more high-quality borrowers into bank loans.

2 Incremental contribution of the paper

- **Relative to delegated monitoring.** Diamond (1984) explains why banks monitor; this paper explains which borrowers demand bank monitoring when reputation is available.
- **Relative to reputation models.** The paper embeds dynamic reputation into the financing choice between monitored and unmonitored debt.
- **New sorting prediction.** Bank-loan borrowers are concentrated in the middle of the credit-quality spectrum, not simply at the bottom.
- **Dynamic graduation.** The paper gives a theoretical foundation for firms starting with bank loans and later accessing public debt markets.
- **Macro channel.** High interest rates or low profitability change bank-loan demand because they change the present value of reputation.
- **Screening versus incentives.** Monitoring may either change behavior or simply identify bad actions. The distinction matters for credit rationing.

3 Model primitives and measurement

3.1 Agents, technologies, and borrower types

All agents are risk neutral. Lenders have funds and can store resources at gross return R . Borrowers need to borrow one unit to operate a project.

- Type G : safe positive-NPV project paying $G > R$.
- Type B : risky negative-NPV project paying B with probability π , with $B > G$ and $\pi B < R$.
- Type BG : can choose the safe project g or the risky project b .

Interpretation. The model separates adverse selection from moral hazard: type uncertainty affects credit terms, while type BG creates the action problem.

3.2 Reputation and public beliefs

The track record Ω_t contains repayments, defaults, and monitoring outcomes. Lenders form posterior beliefs f_t^G , f_t^B , and f_t^{BG} . A higher f_t^B is a worse rating. A perfect track record means no default and no adverse monitoring report.

Interpretation. Credit rating is endogenous public belief, not an exogenous label.

3.3 Debt contracts and monitoring

Borrowers offer a debt face value r_t . If the borrower pays less than r_t , liquidation occurs and yields zero. Monitoring costs C and detects risky action by a BG borrower with probability P :

$$\Pr(m_t = b \mid BG, a_t = b) = P, \quad \Pr(m_t = 0 \mid BG, a_t = b) = 1 - P.$$

If the borrower is G , B , or BG choosing g , monitoring yields no adverse report.

Interpretation. A bank loan is a debt contract that uses costly private monitoring information; direct debt uses only public information.

4 Models and theoretical specifications

4.1 Project choice without monitoring

A BG borrower chooses the safe project if

$$G - r_t + V_{t+1} \geq \pi(B - r_t + V_{t+1}).$$

Equivalently,

$$r_t \leq A_t \equiv V_{t+1} + \frac{G - \pi B}{1 - \pi}.$$

Interpretation. Reputation raises A_t through V_{t+1} : more future rents make current risk shifting less attractive.

4.2 Monitoring that provides incentives

With monitoring, risky action is detected with probability P . A useful incentive condition is

$$G - r_t + V_{t+1} \geq (1 - P)\pi(B - r_t + V_{t+1}),$$

so monitoring can induce safe action when

$$r_t \leq I_t \equiv V_{t+1} + \frac{G - \pi(1 - P)B}{1 - \pi(1 - P)}.$$

Interpretation. Since $I_t > A_t$ when $P > 0$, monitoring can discipline borrowers that reputation alone cannot discipline.

4.3 Screening, credit rationing, and cheap monitoring

If monitoring cannot induce safe action, it may still screen by rejecting borrowers caught choosing b . Screening is weaker than incentive monitoring because it does not change behavior for all borrowers. If monitoring is very cheap and effective, being revealed as BG may carry little stigma because future monitoring can control behavior; monitoring can then undermine its own incentive value.

4.4 Credit-quality regions

High-rated borrowers borrow directly. Middle-rated borrowers use bank loans with incentive monitoring. Lower-rated borrowers may face screening or no credit. This non-monotone sorting is the central model prediction.

5 Identification and theoretical strategy

This is a theoretical paper. Identification comes from equilibrium restrictions that map primitives into debt choice. The paper separates public reputation from private bank monitoring, derives threshold conditions for safe behavior, and then studies how credit ratings and macro conditions move borrowers across direct debt, bank monitoring, screening, and no-credit regions.

5.1 Empirical predictions

- Bank borrowers should be concentrated among middle credit ratings.
- New borrowers should rely more on bank loans and later migrate to public debt.
- High real interest rates or low future profitability should increase the bank-loan share relative to commercial paper.
- New bank-loan quality can improve in bad macro states because safer borrowers newly require monitoring.
- Low-rated borrowers may appear stochastically rationed because banks privately screen them.

5.2 Limitations

The model abstracts from long-term contracts, intermediary incentive constraints, lender reputation, syndication, loan sales, and detailed maturity choice. These abstractions isolate the reputation-monitoring mechanism.

6 Figures from the paper

6.1 Figure 1: Time lines

Read: The upper timeline shows monitoring before loan approval. The lower timeline shows an equivalent covenant structure in which monitoring occurs after the loan is granted and a bad monitoring signal triggers liquidation.

Economic message. The figure clarifies that bank loans are debt contracts plus a costly action-monitoring technology.

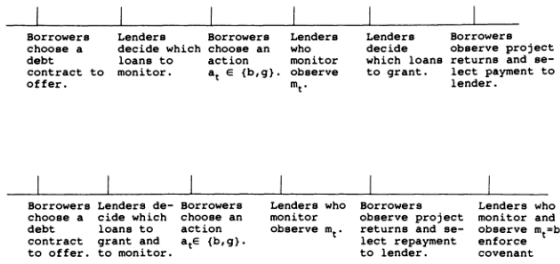


Figure 1: Time lines.

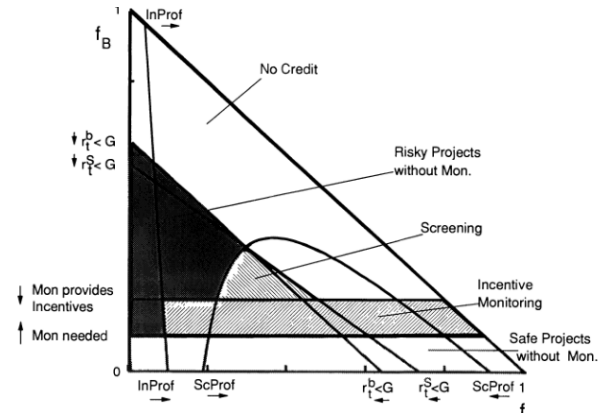


Figure 2: Lending, monitoring, and project choices.

6.2 Figure 2: Lending, monitoring, and project choices at a date $t < T$

Read: The vertical axis is f_B , the posterior probability of the pure bad type; the horizontal axis is f_{BG} , the prevalence of moral hazard. The figure maps regions of direct lending, incentive monitoring, screening, risky unmonitored projects, and no credit.

Economic message. The figure summarizes the core non-monotone sorting result: bank monitoring is most valuable for middle-rated borrowers.

7 Interpretive synthesis

- Banks exist because monitoring can change borrower behavior.
- Banks are not always needed because good reputation can substitute for monitoring.
- The worst borrowers are not necessarily the main bank borrowers because they may have too little reputation to lose.
- Macro downturns can increase bank-loan demand by reducing the value of reputation.
- Monitoring builds reputation, and reputation determines whether monitoring is necessary.

8 Short summary

- The paper studies the choice between bank loans and directly placed debt.
- Borrowers have private types and some face a moral-hazard project choice.
- Reputation disciplines high-rated borrowers.
- Monitoring disciplines middle-rated borrowers.
- Low-rated borrowers may be screened or rationed.
- Borrowers can graduate from bank loans to public debt after building a good record.
- High rates or low future profitability can push more high-rated borrowers into bank loans.

9 Conclusion

9.1 Core conclusion

Bank monitoring is most valuable when reputation alone is insufficient but reputation is still valuable enough that being caught or defaulting is costly.

9.2 Incremental takeaway

The paper connects delegated monitoring, borrower reputation, and debt-market choice. It explains why bank loans are used by middle-rated borrowers, why firms graduate to public debt, and why macro conditions can shift demand for bank monitoring.

9.3 Final takeaway

Bank loan demand is not simply a signal of poor borrower quality. It is a signal that monitoring has marginal incentive value.